

QpiAI™ - Joint AI Training and Certification Program

K-Means Clustering

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Learning with the help
of training examples
without the use of
training labels.

Clustering (K-Means, GMM, etc.)

Manifold Learning (PCA, Kernel PCA, t-SNE
etc.)

Difference Between Clustering and Classification

PARAMETER	CLASSIFICATION	CLUSTERING
Type	Supervised Learning	Unsupervised Learning
Description	Process of classifying the input instances based on their corresponding class labels	Grouping the instances based on their similarity without the help of class labels
The necessity of Training and Testing Dataset	It has labels so there is a need for training and testing dataset for verifying the model created	There is no need for training and testing dataset
Complexity	More complex as compared to clustering	Less complex as compared to the classification
Example Algorithms	Logistic regression, Support Vector Machines, etc.	K-Means Clustering Algorithm, Gaussian Clustering Algorithm, etc.

What is Clustering?

- Clustering aims to partition the data into distinct groups so that the observations within each group are homogenous and quite similar to each other. We understand with a simple example.
- A bank wants to give credit card offers to its customers. Currently, they look at the details of each customer and based on this information, decide which offer should be given to which customer.
- The bank can potentially have millions of customers. And making decision individually for each and every customer would be a strenuous task.
- So what can the bank do? One option is to segment its customers into different groups. For instance, the bank can group the customers based on their income, occupation, age group etc.



Real World Use Cases Of Clustering

Store Segmentation for a Cereal Brand

- Zip code
- Competitor Product Sold
- Age Group of People
- Income Group of People
- Product Price bin for competitor
- Ethnicity
- Product group category of competitor

Document Clustering

- Key-words
- Author
- Context: Date, Location, Time

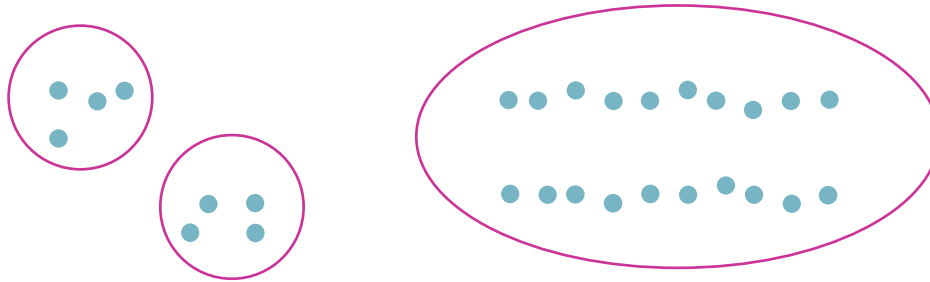
Customer Segmentation for all products to see whether user will renew the contract or not?

- Median usage of days
- Median of session numbers
- Median of session length
- Fluctuation of product use
- Patterns of session frequency and length
- Usage recency before earlier renewals

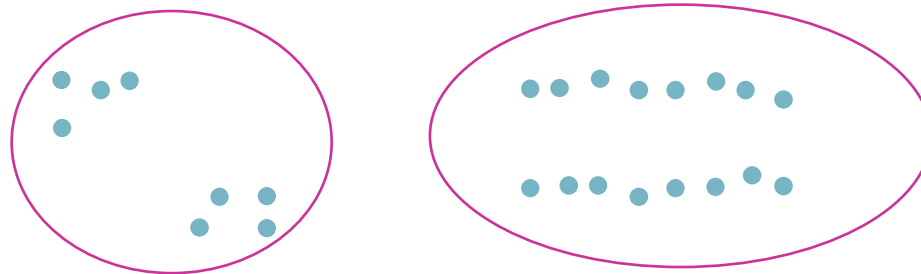
Basic idea: Group together similar instances

Example: 2D point patterns

Case 1



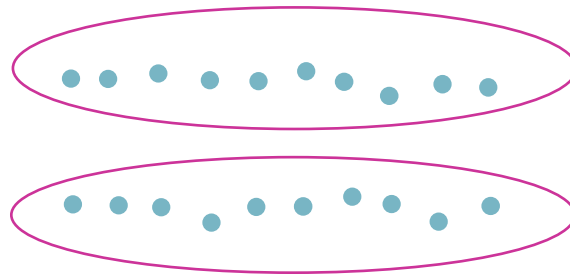
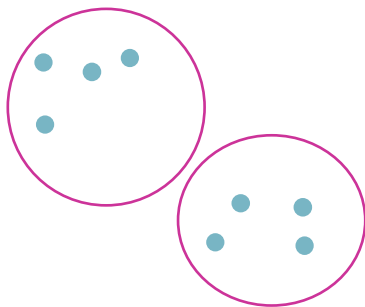
Case 2



Basic idea: Group together similar instances

Example: 2D point patterns

Case 3



What could “similar” mean?

- One option: small Euclidean distance (squared)

$$\text{dist}(\vec{x}, \vec{y}) = ||\vec{x} - \vec{y}||^2$$

- Clustering results are crucially dependent on the measure of similarity (or distance) between “points” to be clustered and the domain knowledge of the data points being considered.

Introduction to K-Means Clustering

The points within a cluster should be similar to each other. So, our aim here is to minimize the distance between the points within a cluster.

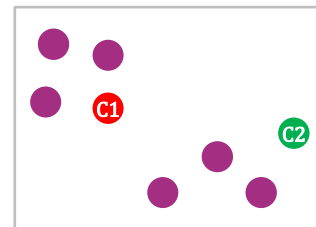
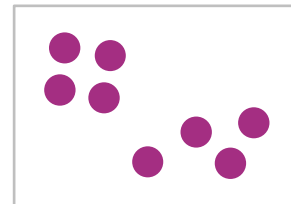
Let's now take an example to understand how K-Means actually works: We have these 8 points and we want to apply K-Means to create clusters for these points. Here's how we can do it.

Step 1: Choose the number of clusters k

The first step in K-Means is to pick the number of clusters, k.

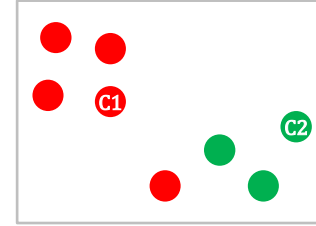
Step 2: Select k random points from the data as centroids

Next, we randomly select the centroid for each cluster. Let's say we want to have 2 clusters, so k is equal to 2 here. We then randomly select the centroid:



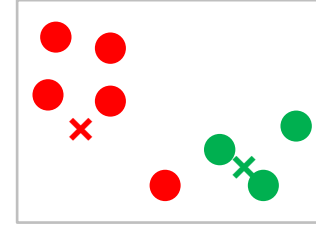
Step 3: Assign all the points to the closest cluster centroid

Once we have initialized the centroids, we assign each point to the closest cluster centroid:



Step 4: Recompute the centroids of newly formed clusters

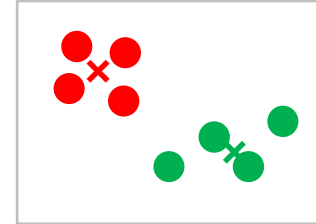
Now, once we have assigned all of the points to either cluster, the next step is to compute the centroids of newly formed clusters:



Step 5: Repeat steps 3 and 4

We then repeat steps 3 and 4:

The step of computing the centroid and assigning all the points to the cluster based on their distance from the centroid is a single iteration.



Problem Definition

Input: A set of points $x_1, \dots, x_n \in \mathbb{R}^d$; an integer k denoting the number of clusters

Output: A partition of the points into clusters $C_1, \dots, C_k$ along with a center μ_j for each cluster, so as to minimize

$$\sum_{j=1}^k \sum_{i \in C_j} \|x_i - \mu_j\|^2$$

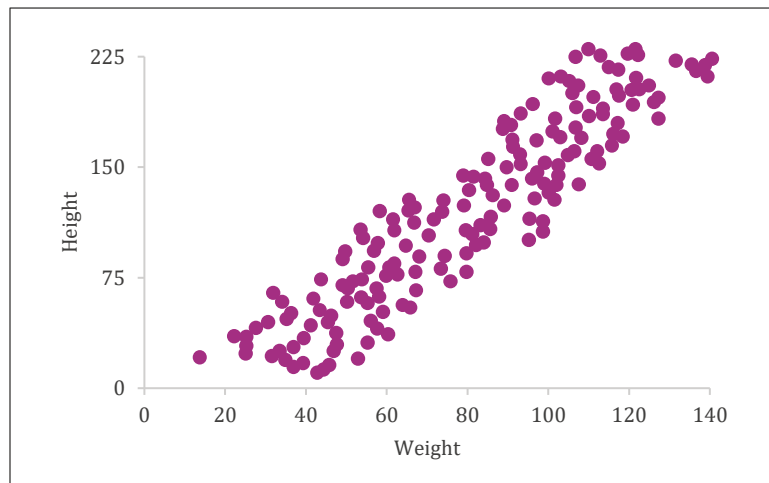
The K-Means cost function can equivalently be rewritten as

$$\sum_{j=1}^k \frac{1}{2|C_j|} \sum_{i, i' \in C_j} \|x_i - \mu_{i'}\|^2$$

Example

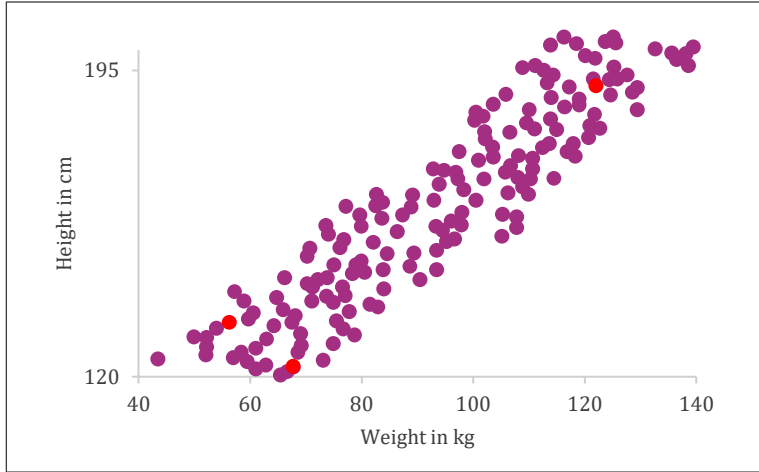
Height	Weight
166	68.30679
178	55.79946
169	67.02459
173	78.27659
177	58.50331
182	110.37332
186	102.73013
168	83.79284
121	44.04066
126	45.36331

- We are given a dataset of Height and Weight of 170 people
- We would like to group the people such that we can assign T-shirts to the group of people
- We have 3 sizes of T-shirt available with us S,M,L

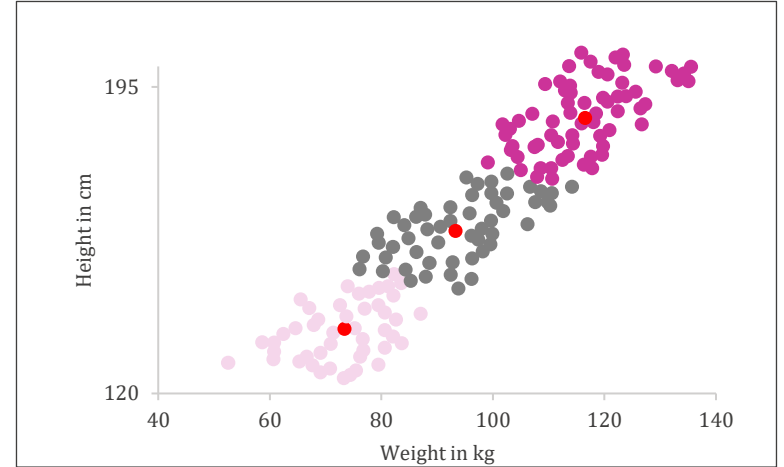


Example

- Assigning random centers and starting the iterations



- We have 3 clusters with us and on the basis of these clusters we can easily assign T-shirts to the group



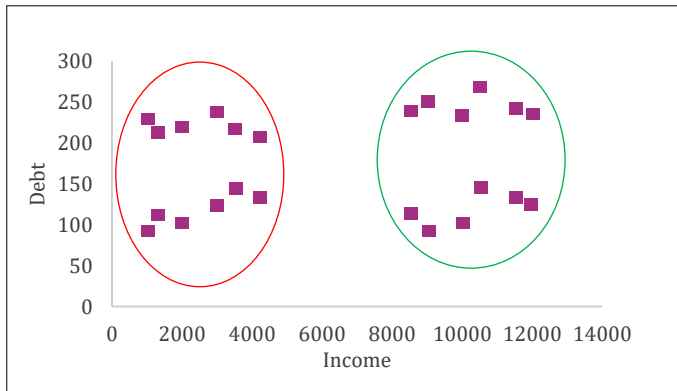
Stopping Criterion for K-Means Clustering

There are essentially three stopping criteria that can be adopted to stop the K-Means algorithm:

1. Centroids of newly formed clusters do not change within a specified tolerance of 2 consecutive iterations
2. Points remain in the same cluster
3. Maximum number of iterations is reached

How to Choose the Right Number of Clusters in K-Means Clustering?

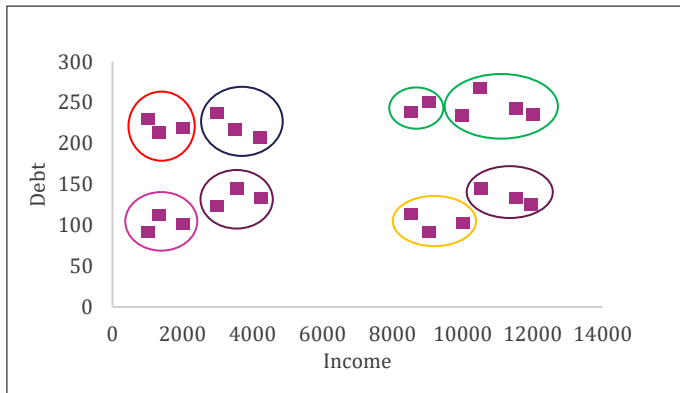
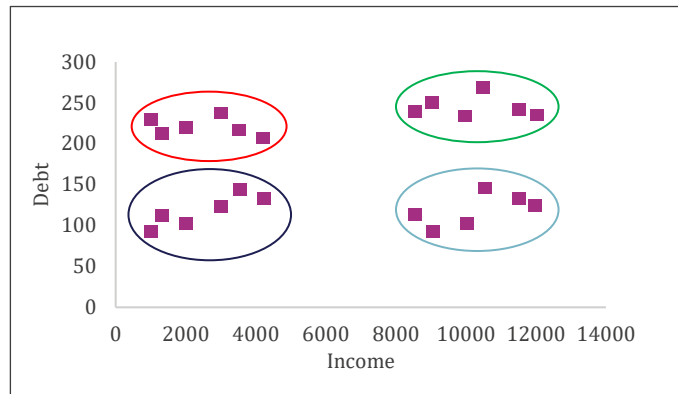
Let's take the customer segmentation example which we saw earlier. To recap, the bank wants to segment its customers and assign them cards:



All the customers with low income are in one cluster whereas the customers with high income are in the second cluster

How to Choose the Right Number of Clusters in K-Means Clustering?

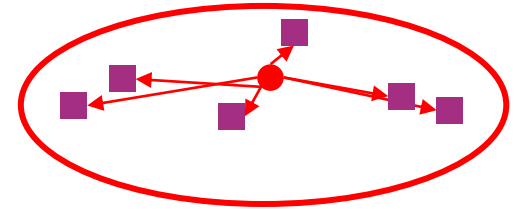
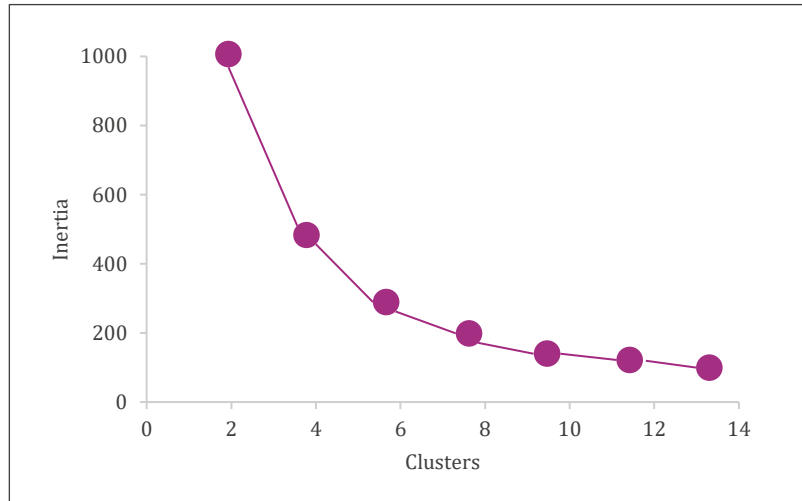
One cluster might represent customers who have low income and low debt, other cluster is where customers have high income and high debt, and so on



The maximum possible number of clusters will be equal to the number of observations in the dataset!

Elbow Method

- **Inertia metric:** calculates the sum of distances of all the points within a cluster from the centroid of that cluster.
- We calculate this for all the clusters and the final inertia value is the sum of all these distances. This distance within the clusters is known as **intracluster distance**. So, inertia gives us the sum of **intracluster distances**:



Intra cluster distance

Disadvantages of K-Means

1

Sensitive to Outliers.

2

Sensitive to the choice of the initial seeds (centroids).

3

Need not converge to the global optimum.

4

May not always be able to cluster properly.

Disadvantages of K-Means

1

Sensitive to Outliers.



(A): Undesirable clusters

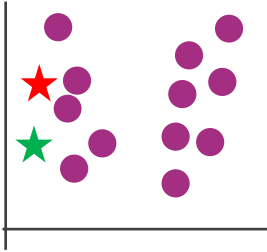


(B): Ideal Clusters

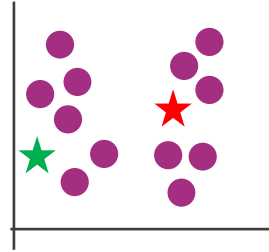
Disadvantages of K-Means

2

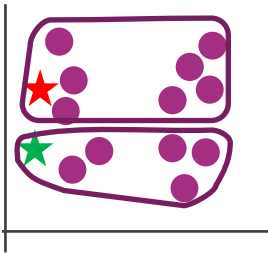
Sensitive to the choice of initial seeds



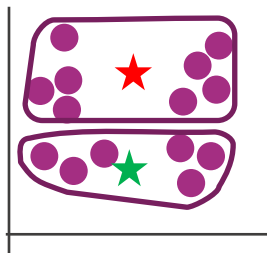
Random selection of seeds (centroids)



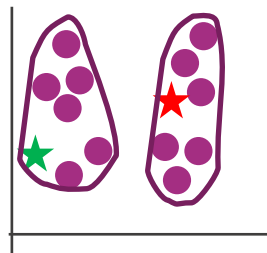
Random selection of seeds (centroids)



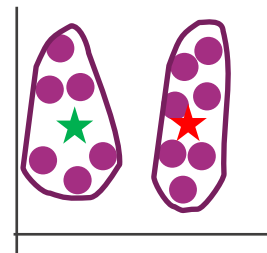
Iteration 1



Iteration 2



Iteration 1

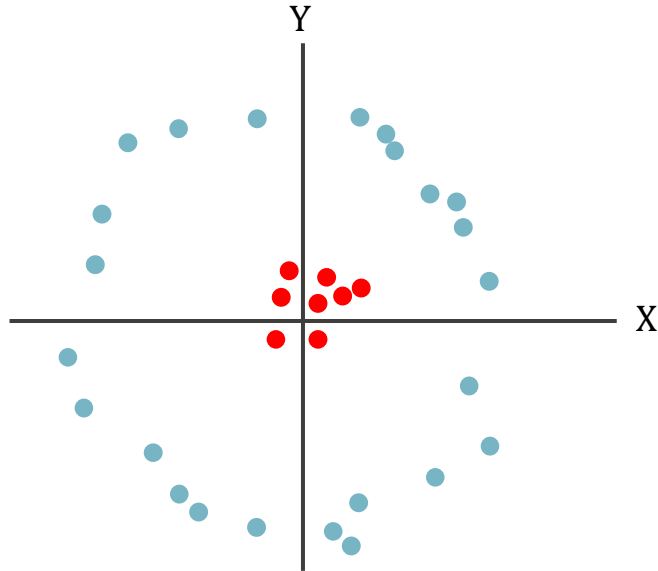


Iteration 2

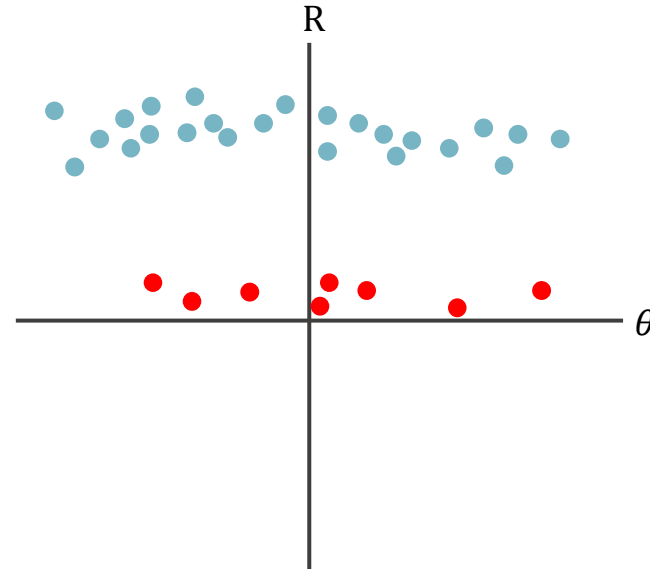
Disadvantages of K-Means

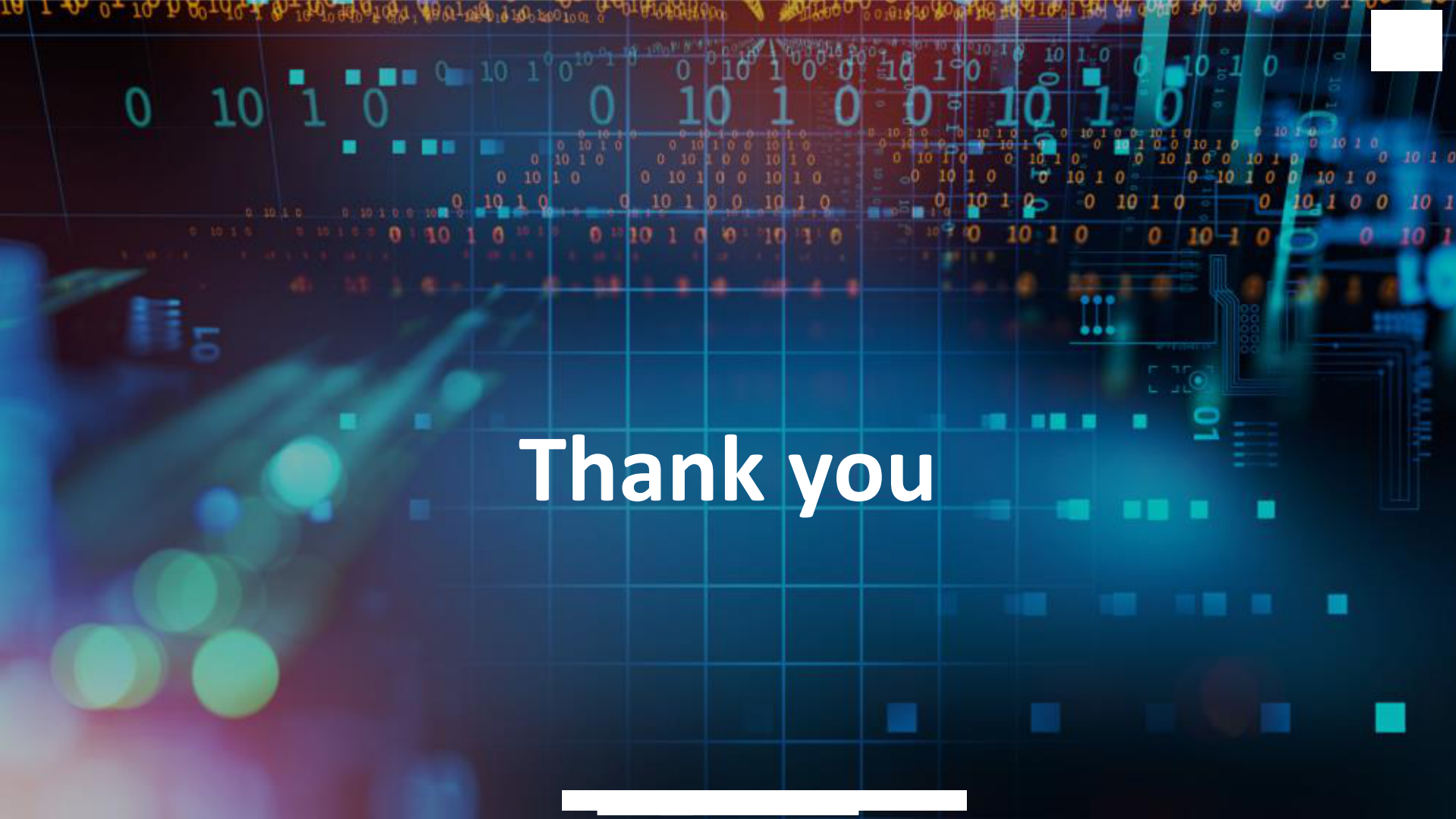
4

May not cluster properly



Changing the features
(distance function) can help





Thank you